



ENGINEERING MATHEMATICS - II

UE20MA151

Dr. SIVASANKARI. V
Department of Science & Humanities

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Unit 4 : Inverse Laplace Transform

Session : 10

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Laplace Transform can also be used to

- a) Solve ODE with Variable Coefficients
- b) Solve Simultaneous ODE

1) Using Laplace Transforms,

Solve $ty'' + 2y' + ty = \cos t$, $y(0) = 1$, $y'(0) = 0$.

Taking Laplace Transforms of both sides,

$$L[ty''(t)] + 2L[y'(t)] + L[ty(t)] = L[\cos t]$$

$$-\frac{d\{L[y''(t)]\}}{ds} + 2L[y'(t)] - \frac{d\{L[y(t)]\}}{ds} = L[\cos t]$$

$$-\frac{d}{ds}\{s^2L[y(t)] - sy(0) - y'(0)\} + 2\{sL[y(t)] - y(0)\} - \frac{d}{ds}\{L[y(t)]\} = \frac{s}{s^2 + 1}$$

$$-\frac{d}{ds}\{s^2 L[y(t)] - s\} + 2\{sL[y(t)] - 1\} - \frac{d}{ds}\{L[y(t)]\} = \frac{s}{s^2 + 1}$$

$$-\{s^2\} \frac{d}{ds} L[y(t)] - 2sL[y(t)] + 1 + 2sL[y(t)] - 2 - \frac{d}{ds}\{L[y(t)]\} = \frac{s}{s^2 + 1}$$

$$\{-s^2 - 1\} \frac{d}{ds} L[y(t)] - 1 = \frac{s}{s^2 + 1}$$

$$\{s^2 + 1\} \frac{d}{ds} L[y(t)] + 1 = \frac{-s}{s^2 + 1}$$

$$\{s^2 + 1\} \frac{d}{ds} L[y(t)] = \frac{-s}{s^2 + 1} - 1$$

$$\{s^2 + 1\} \frac{d}{ds} L[y(t)] = \frac{-s}{s^2 + 1} - 1$$

$$\frac{d}{ds} L[y(t)] = \frac{-s}{(s^2 + 1)^2} - \frac{1}{s^2 + 1}$$

$$ty(t) = L^{-1} \left\{ \frac{-s}{(s^2 + 1)^2} \right\} - L^{-1} \left\{ \frac{1}{s^2 + 1} \right\}$$

$$ty(t) = \frac{1}{2} tsint + sint$$

$$y(t) = \frac{1}{2} sint + \frac{\sin t}{t}$$

$$L\{tf(t)\} = -\frac{d}{ds} L\{f(t)\}$$
$$L^{-1} \left\{ \frac{a}{(s^2 + a^2)^2} \right\} = \frac{tsinat}{2a}$$

2) Using Laplace Transforms, Solve the simultaneous equations

$$\frac{dx}{dt} - 2y = \cos 2t, \quad \frac{dy}{dt} + 2x = \sin 2t, \quad x(0) = 1, y(0) = 0.$$

$$\text{➤ } L[x'(t)] - 2L[y(t)] = L[\cos 2t]$$

$$\text{➤ } \{sL[x(t)] - x(0)\} - 2L[y(t)] = L[\cos 2t]$$

$$\text{➤ } \{sL[x(t)] - 1\} - 2L[y(t)] = \frac{s}{s^2+4}$$

$$\text{➤ } sL[x(t)] - 2L[y(t)] = \frac{s}{s^2+4} + 1$$

$$\text{➤ } L[y'(t)] + 2L[x(t)] = L[\sin 2t]$$

$$\text{➤ } \{sL[y(t)] - y(0)\} + 2L[x(t)] = L[\sin 2t]$$

$$\text{➤ } \{sL[y(t)] - 0\} + 2L[x(t)] = \frac{2}{s^2+4}$$

$$\text{➤ } sL[y(t)] + 2L[x(t)] = \frac{2}{s^2+4}$$

Consider

$$\text{➤ } sL[x(t)] - 2L[y(t)] = \frac{s}{s^2+4} + 1 \dots\dots\dots(1) \quad X(-2)$$

$$\text{➤ } 2L[x(t)] + sL[y(t)] = \frac{2}{s^2+4} \dots\dots\dots(2) \quad X(s)$$

$$\text{➤ } -2sL[x(t)] + 4L[y(t)] = -\frac{2s}{s^2+4} - 2 \dots\dots\dots(3)$$

$$\text{➤ } 2sL[x(t)] + s^2L[y(t)] = \frac{2s}{s^2+4} \dots\dots\dots(4)$$

Adding (3) & (4)

$$(s^2+4)L[y(t)] = \frac{2s}{s^2+4} - \frac{2s}{s^2+4} - 2$$

$$\Rightarrow L[y(t)] = -\frac{2}{s^2+4}$$

$$\Rightarrow y(t) = -\sin 2t$$

Substituting this in (2)

$$2L[x(t)] + sL[-\sin 2t] = \frac{2}{s^2 + 4}$$

$$\Rightarrow 2L[x(t)] - \frac{2s}{s^2 + 4} = \frac{2}{s^2 + 4}$$

$$\Rightarrow L[x(t)] = \frac{s}{s^2 + 4} + \frac{1}{s^2 + 4}$$

$$\Rightarrow x(t) = L^{-1} \left[\frac{s}{s^2 + 4} \right] + L^{-1} \left[\frac{1}{s^2 + 4} \right]$$

$$\Rightarrow x(t) = \cos 2t + \frac{\sin 2t}{2}$$

Therefore solution is

$$x(t) = \cos 2t + \frac{\sin 2t}{2} \quad ; \quad y(t) = -\sin 2t$$

3) The ODE below models the behaviour of a time varying current $i(t)$ in a circuit $\frac{di}{dt} + i = v(t)$ where $v(t) = \begin{cases} v_0, & 0 \leq t \leq 1 \\ 0, & t \geq 1 \end{cases}$ is a source voltage and $i(0) = v_0$

- Write as a single line expression using step functions.
- Then determine $i(t)$

Solution : i. Let $v(t) = v_0 - v_0 u(t - 1)$

$$\text{Then } \frac{di}{dt} + i = v(t)$$

$$\Rightarrow \frac{di}{dt} + i = v_0 - v_0 u(t - 1)$$

Taking Laplace Transform on both sides

$$L\left[\frac{di}{dt}\right] + L[i] = L[v_0] - L[v_0 u(t - 1)]$$

$$\Rightarrow \{sL[i] - i(0)\} + L[i] = \frac{v_0}{s} - \frac{v_0 e^{-s}}{s}$$

$$\Rightarrow (s + 1)L[i] - v_0 = \frac{v_0}{s} - \frac{v_0 e^{-s}}{s}$$

$$\Rightarrow (s + 1)L[i] = v_0 + \frac{v_0}{s} - \frac{v_0 e^{-s}}{s}$$

$$\Rightarrow L[i] = v_0 \left[\frac{1}{s+1} + \frac{1}{s(s+1)} - \frac{e^{-s}}{s(s+1)} \right]$$

$$\Rightarrow L[i] = v_0 \left[\frac{1}{s+1} + \frac{s+1-s}{s(s+1)} - \frac{e^{-s}(s+1-s)}{s(s+1)} \right]$$

$$\Rightarrow L[i] = v_0 \left[\frac{1}{s+1} + \frac{1}{s} - \frac{1}{s+1} - \frac{e^{-s}}{s} + \frac{e^{-s}}{s+1} \right]$$

$$\Rightarrow L[i] = v_0 \left[\frac{1}{s} - \frac{e^{-s}}{s} + \frac{e^{-s}}{s+1} \right]$$

$$\Rightarrow i = v_0 L^{-1} \left\{ \frac{1}{s} - e^{-s} \left[\frac{1}{s} - \frac{1}{s+1} \right] \right\}$$

$$\Rightarrow i = v_0 - v_0 [1 - e^{-(t-1)}] u(t-1)$$

$$\text{since } L^{-1}[e^{-as}F(s)] = f(t-a)u(t-a)$$



THANK YOU

Dr. SIVASANKARI. V

Department of Science & Humanities

sivasankariv@pes.edu